Strategic Production of Persistent Information

Research summary accompanying the working paper

Lucas Sneller Graduate Student, Quantitative Methods in the Social Sciences
Columbia University September 10, 2026

The connected question

Can decentralized incentives eliminate consequential public artifacts as a population grows, even when a coordinated policy would sustain them and improve average welfare? The working paper answers yes under a specific allocation-and-readiness institution. The contribution is the connection to *which shared information environment survives*, not a new general free-riding or common-noise principle.

One architecture; two fixed information channels

Agent i has state $Y_i = (X_i, C_i) \in \{0,1\} \times \{U, E, A\}$: a binary task coordinate and an unexposed, exposed, or adopter communication state. Write $\mu^N = N^{-1} \sum_i \delta_{Y_i}$ and $a(\mu) = \mu(0, A) + \mu(1, A)$. A shared board $B \geq 0$ stores past deposits and decays at rate κB. Communication changes as

$$U \xrightarrow{\rho B} E, \qquad E \xrightarrow{\alpha} A, \qquad E \xrightarrow{\nu} U, \qquad A \xrightarrow{\delta} U.$$

Tasks switch $0 \to 1$ at rate c_0 for nonadopters and c_1 for adopters, and reset $1 \to 0$ at rate r. Deposits change only the board. Labeled states and the board are observed in the game; the task coordinate is not hidden information.

Channel	Per-adopter rate	Board increment	Large-population effect
Routine	λ_w	η/N	Deterministic accumulation
Rare artifact	$\lambda_M u_i/N$	$h > 0$	Policy-dependent jumps

Routine deposits are an uncontrolled background. The strategic paper controls only rare-artifact production. With eligible feedback θ, the specified fixed-policy limit is

$$\dot{\mu}_t = \mu_t Q(B_t), \qquad dB_t = \{-\kappa B_t + w\eta\lambda_w a(\mu_t)\}\, dt + h\, dJ_t,$$

where $w = 1$ enables routine deposits and J thins a common Poisson proposal measure at intensity $\lambda_M \sum_{y:C(y)=A} \mu_{t-}(y)\theta(t, y, \mu_{t-}, B_{t-})$. The board jumps; the population law stays continuous. Under equilibrium the rare channel disappears. [Paper, §4]

What independence means. In the deterministic limit, finitely many tagged paths have an ordinary product law. Under the productive fixed policy, they are independent conditional on the common proposal measure; averaging over that common history produces dependence. The population estimates assume bounded initial boards, convergence to deterministic initial data in L^p for some $p > 2$, and independence from future clocks. Tagged statements also require exchangeable chaotic initialization. [Paper, §4; App. A]

The substantive institution

At rate $\lambda_M \bar{u}$, a fixed opportunity stream is allocated uniformly among N players. An eligible recipient succeeds with probability $u_i/\bar{u}$. Opportunities cannot be reassigned, and ineligible allocations expire. Agents choose and pay for readiness *before* assignment. Effort units, cap, costs, and per-person benefits remain fixed as N grows. These are separate assumptions, not consequences of capacity. Their relevance to multi-agent AI systems is unvalidated. [Paper, §§1–2, 6]

Independent working paper, Public v1. Developed independently with AI assistance; not peer reviewed. This summary accompanies the full working paper. The university affiliation identifies the author's student status.

Incentives, welfare, and limiting dependence

Only adopters choose costly readiness $u_i \in [0, \bar{u}]$. Effort controls rare production, with no additional direct private benefit:

$$\Lambda_N(t) = \frac{\lambda_M}{N} \sum_{i=1}^{N} \mathbf{1}_{\{C_i(t-)=A\}} u_i(t), \qquad c(u) = \ell u + \frac{\gamma}{2} u^2, \quad \ell \geq 0,\ \gamma > 0.$$

Payoffs over $[0, T]$ are expected discounted task rewards minus readiness costs, with $0 \leq g_i \leq R$, discount $\beta \geq 0$, and zero terminal payoff. Welfare is their per-person average. [Paper, §2.1]

Why both individual effort and aggregate intensity vanish. Let V_i^N be a no-transfer equilibrium continuation value and $\Delta_h V_i^N = V_i^N(s^h) - V_i^N(s)$, where s^h adds one artifact. Zero effort is admissible, so

$$0 \leq V_i^N \leq K_T := R \int_0^T e^{-\beta t} dt, \qquad u_i^{\mathrm{NE}} = \mathrm{clip}_{[0,\bar{u}]} \left(\frac{(\lambda_M/N) \Delta_h V_i^N - \ell}{\gamma} \right)$$

for eligible players. Thus

$$u_i^{\mathrm{NE}} \leq \frac{[\lambda_M K_T/N - \ell]_+}{\gamma}; \qquad \ell = 0 \implies \Lambda_N^{\mathrm{NE}} \leq \frac{\lambda_M^2 K_T}{\gamma N}.$$

The continuation difference includes future task effects and rivals' feedback. Benefits need not be $O(N^{-1})$: this factor enters through individual discovery technology. Positive ℓ gives exact shutdown for sufficiently large N. [Paper, Cor. 2.2]

An equilibrium exists before taking the limit. For every finite N, T, a pure continuous Markov-perfect equilibrium is verified against admissible predictable unilateral deviations. Uniqueness is confined to the *bounded jointly continuous mild-HJB class*. The proof adds bounded board jumps to the rare-disabled semigroup, solves backward on short slabs, and continues using the verified reward bound. A potential martingale interlaces proposals and predictable acceptance; rivals retain feedback at the deviating state. No board derivatives or global supremum-norm strong continuity are required. [Paper, Thm. 2.1; App. A.1]

Equilibrium loses the rare channel. On every fixed horizon, $\mathbb{P}(J_T^{N,\mathrm{NE}} > 0) = O(N^{-1})$. A shared-clock coupling agrees with the rare-disabled system until the first artifact, giving $O(N^{-1})$ welfare error and, with bounded initial boards, expected uniform state error. The limiting environment is therefore deterministic, possibly retaining routine drift, with ordinary tagged independence under the initialization assumptions. [Paper, Prop. 4.1]

Coordination remains strictly productive in a concrete region. Take rare-only production, $B_0 = 0$, reward $g_i = RX_i$, positive $R, \rho, \alpha, h, \lambda_M, \bar{u}$, and $c_1 > c_0$. Initially a positive fraction are adopters and a positive fraction are incomplete and unexposed, uniformly in N. For quadratic costs, a sufficiently small fixed effort $\varepsilon > 0$ by adopters until $t_0 < T$ satisfies

$$W_N^{\varepsilon} - W_N^0 = \varepsilon A_N + O(\varepsilon^2), \quad A_N \geq A_* > 0, \qquad \liminf_{N \to \infty} (W_N^{\mathrm{SP}} - W_N^{\mathrm{NE}}) \geq \delta_* > 0.$$

The bounds are uniform in admissible N; small positive ℓ is also allowed. This seeded, zero-board, rare-only result does not establish productivity for arbitrary initial conditions or hybrid backgrounds. [Paper, Thms. 3.1, 3.3]

The retained common jumps affect useful outcomes. That specified early-effort policy has a nondegenerate common-jump limit. For exchangeable limiting tags with discounted task rewards $H_i = \int_0^T e^{-\beta t} RX_i(t) dt$, conditional independence gives

$$\mathrm{Cov}(H_1, H_2) = \mathrm{Var}(\mathbb{E}[H_1 \mid \Pi^0]) > 0.$$

For bounded, spatially Lipschitz feedback, convergence is quantitative in rare-only mode but only qualitative in the controlled hybrid model. Neither optimal-value nor optimal-control convergence is established. Productivity is a claim about artifacts, not randomness itself. [Paper, Thm. 4.2; §§4.3–4.4]

A solved benchmark and the limits of creator credit

The two-state calculation. An artifact is absent or present. When absent, discovery intensity is $q_N = a_N \sum_i u_i$; when present, each player receives flow benefit $v > 0$ and the artifact depreciates at rate $d \geq 0$. For an infinite horizon with fixed $\beta > 0$, put $k = \beta + d$. Let D^{SP} be the planner's per-person value advantage of a present artifact. With $a_N = \lambda/N$, quadratic costs, and an inactive cap,

$$u_N^{\mathrm{NE}} \sim \frac{\lambda v}{\gamma k N}, \qquad q_N^{\mathrm{NE}} \sim \frac{\lambda^2 v}{\gamma k N}, \qquad kD^{\mathrm{SP}} + \frac{\lambda^2}{2\gamma}(D^{\mathrm{SP}})^2 = v.$$

The planner chooses $u^{\mathrm{SP}} = \lambda D^{\mathrm{SP}}/\gamma$, sustaining a positive, N-independent hazard. Capped and linear-cost versions are also solved. This explicit benchmark is distinct from the analytically characterized six-state game: no numerical six-state Nash equilibrium was computed. [Paper, §§1.1–1.4, 7]

Capacity alone does not produce collapse. With exogenous access probabilities π_i, the same capacity gives intensity $\lambda \sum_i \pi_i u_i$. Assigning all access to one designated creator ($\pi_1 = 1$) yields an N-independent private problem, active when $\lambda v > k\ell$ and the cap is positive. This is a counterfactual to dispersed allocation, not a model of endogenous hierarchy or efficient delegation. [Paper, Prop. 1.2]

The central scaling boundary is $s = 1/2$, not uniquely $s = 1$. For the two-state benchmark with $a_N = \lambda N^{-s}$, $s \geq 0$, quadratic costs, fixed positive $\lambda, v, \gamma, \beta, \bar{u}$, and fixed $d \geq 0$, the equilibrium absent-state hazard satisfies

$$q_N^{\mathrm{NE}} \sim \begin{cases} \lambda\sqrt{v/\gamma}\, N^{1/2-s}, & s < 1/2, \\ \lambda^2 D_c/\gamma, & s = 1/2, \\ \lambda^2 v(\gamma k)^{-1} N^{1-2s}, & s > 1/2, \end{cases} \qquad kD_c + \frac{\lambda^2}{\gamma} D_c^2 = v.$$

Every fixed positive cap is eventually nonbinding. For $\ell > 0$, every $s > 0$ instead eventually shuts down; at $s = 0$, activity requires $\lambda v > k\ell$. The potentially diverging hazard is conditional on absence: stationary renewal frequency is $dq_N/(d + q_N) \leq d$. These are not six-state scaling theorems or joint long-horizon/vanishing-discount results. Finite-horizon artifact disappearance does not imply impossibility of eventual discovery at each finite N. [Paper, Prop. 1.3; §1.3]

Exact implementation in the benchmark. Pay the creator Np and charge every player p per success, including the creator. Net transfers balance exactly. Setting $p = D^{\mathrm{SP}}$ implements planner effort as a symmetric stationary Markov equilibrium in the two-state model, under risk neutrality, reliable attribution, enforceable levies, and commitment. This is an implementation result, not unrestricted uniqueness. [Paper, Prop. 5.1]

Why the naive six-state lift fails. For an interior planner target u^*, let U_i^p be player i's actual continuation value under the target and transfer rule. Incentive compatibility requires

$$\Delta_h U_i^p + (N-1)p = N\Delta_h W_N^{\mathrm{SP}}.$$

Taking $p = \Delta_h W_N^{\mathrm{SP}}$ therefore requires an *individual* continuation identity that budget balance supplies only on average. In the two-agent counterexample, a completed adopter's target cost and expected transfer cancel, leaving $\Delta_h U_1^p = 0$. Halving the target effort over a sufficiently short positive interval, then reverting, produces initial gain rate $\gamma(u^*)^2/8 > 0$. The paper proves profitable deviation and nonimplementation, not a complete dynamic best-response policy. [Paper, Prop. 5.2; §5.4]

Large creator prizes complicate limiting payoffs. Under the implemented two-state policy on a fixed horizon, a tagged creator's success probability is $O(N^{-1})$ but its gross prize is $O(N)$. With positive expected discovery count, gross payments tend to zero in probability while retaining a positive expectation: they are not uniformly integrable. State-path convergence alone cannot justify convergence of expected transfers. [Paper, §5.2]

Numerical evidence and its limits

The six-state computations evaluate specified policies, including eligible effort 0.25 until time 1.25 on a horizon of 6. The setting is rare-only, with zero board, 20% initial adopters and 80% incomplete unexposed players; all tasks begin incomplete. Each displayed cell uses 160 independent histories. The full paper specifies the remaining rates, cost parameters, seed scheme and integration methods. [Paper, §7]

Population	Welfare above exact zero-effort baseline (95% interval)
60	0.1147 [0.0469, 0.1825]
240	0.1947 [0.1184, 0.2710]
960	0.1240 [0.0618, 0.1863]
Fixed-policy limit	0.0998 [0.0442, 0.1554]

These are per-capita discounted net rewards, with Student intervals across histories. The exact baseline is approximately 1.15065544. The analytic certificate is much smaller: it certifies an effort about 3.70×10^{-7} and a gain about 1.78×10^{-10}. The theorem proves that some fixed positive improvement exists; the larger illustrated improvement at effort 0.25 is simulation evidence. An imposed $0.25/N$ effort diagnostic is not a computed six-state Nash equilibrium.

For this release, tests and deterministic coefficient calculations were run afresh and saved aggregates were rechecked. The displayed simulation histories were retained, not rerun. Code and raw records remain local; this public release supplies the manuscript and summary PDFs, not a public replication package.

Interpretation and a bounded next question

The analysis is a conditional institutional comparison. No evidence here establishes the opportunity allocation or readiness timing for multi-agent AI systems. Known productivity is not Bayesian learning. The model does not establish endogenous hierarchy, efficient specialization, a general mechanism for heterogeneous states, or convergence of optimal controls and values. Its board omits content and artifact-specific exposure.

The next question is how reassignment of unused opportunities changes the comparison when capacity and task transitions are held fixed. That change would test the role of the access institution directly; it remains unsolved in this version. The separate nonstrategic probability manuscript supplies background, while the present public release is the strategic working paper.

Related work

Team discovery and durable public goods already exhibit incentive wedges [2, 1]. Conditional common-Poisson limits provide the probability context [3]. The paper's result connects these questions under its specified production institution; the source comparisons do not establish priority or empirical validity.

Selected references

[1] Marco Battaglini, Salvatore Nunnari, and Thomas R. Palfrey. Dynamic free riding with irreversible investments. *American Economic Review*, 104(9):2858–2871, 2014. doi: 10.1257/aer.104.9.2858. URL https://snunnari.github.io/bnp_dynfreerider_aer.pdf.

[2] Alessandro Bonatti and Johannes Hörner. Collaborating. *American Economic Review*, 101(2):632–663, 2011. doi: 10.1257/aer.101.2.632. URL https://bpb-us-w2.wpmucdn.com/campuspress.yale.edu/dist/3/352/files/2012/02/65916.pdf.

[3] Daniel Hernández-Hernández and Joshué Helí Ricalde-Guerrero. Conditional McKean–Vlasov differential equations with common Poissonian noise: Propagation of chaos. *Stochastics and Dynamics*, 25(7–8):2550033, 2025. doi: 10.1142/S0219493725500339. URL https://arxiv.org/abs/2308.11564. Article 2550033; author version arXiv:2308.11564v3.